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Master Thesis
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# Turning Patent Data into Spatial Intelligence: A Feasibility Study on Linguistic and GIS Data Integration for Water Risk Analysis

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**Abstract**

Water Distribution Networks (WDNs) globally face severe efficiency challenges, losing over 126 billion cubic meters of treated water annually to leakage and structural bursts. In Italy, where average distribution loss reaches 37.9% and over 30% of active conduits exceed their design life expectancy. Because monitoring systems are managed independently by localized, province or city-based service providers, utilities suffer from significant data duplication and isolated control protocols that remain unintegrated with public open-data repositories. To help overcome these operational redundancies and eliminate duplicated measurement efforts, this thesis proposes a spatial-causal data fusion framework that integrates open geospatial data on environmental factors with patent intelligence on technical infrastructural features, systematically evaluating failure risk chains without relying on proprietary or restricted pipeline parameters.

Applied across the Province of Firenze (Toscana, Italy) as a regional proof of concept, the methodology establishes an integrated dual-stream architecture. In the geospatial stream, disparate public environmental, hydro-meteorological, and socio-economic datasets including DTM slope, Sentinel-1 InSAR ground displacement rates, precipitation totals, landslide inventories, and seismic hazards are harmonized onto a uniform Uber H3 hexagonal spatial grid at Resolution 10 ($\sim$ 1-hectare cell resolution), where spatial dependence and compound stressor interaction zones are identified using Standard Deviational Ellipses alongside global and bivariate Local Indicators of Spatial Association (LISA / Moran's $I$). Concurrently, in the patent knowledge stream, a patent corpus of more than 17 thousand patents has been automatically analysed. The patent retrieval phase was supported by Errequadro, a leading patent intelligence company in Pisa. The patent process regards a four-tier SKOS/RDF Pipeline Risk Ontology to extract sentence-level structured causal triplets linking environmental exposure variables to specific technological failure modes.Sentence transformer embeddings then semantically bridge localized spatial activation zones with patent-derived causal mechanisms to identify potential failure triggers. Rather than deterministically proving the presence of active pipe defects a limitation imposed by missing water infrastructure data, the framework highlights vulnerable locations where structural issues may arise, transforming open environmental covariates into actionable engineering risk intelligence under conditions of severe data opacity.

**Key Terms**

*Term 1* — Water Distribution Networks.
*Term 2* — Spatial Causal Data Fusion.
*Term 3* — Uber H3 Hexagonal Grid.
*Term 4* — Environmental Risk Indicators.
*Term 5* — Patent Intelligence.
*Term 6* — Patent Mining.
*Term 7* — Pipeline Failure Analysis.
*Term 8* — Local Indicators of Spatial Association.

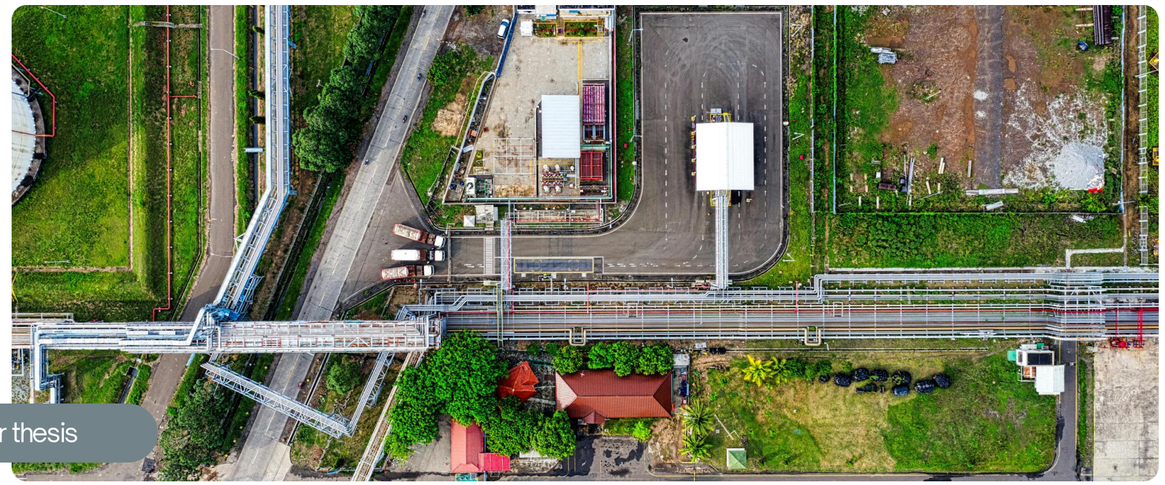
# *Turning Patent Data into Spatial Intelligence:*

# *A Feasibility Study on Linguistic and GIS Data Integration for Water Risk Analysis*

Transforming multimodal open data into decision making


Candidate

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Master thesis

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Context

*How can we define the vulnerabilities of a Water distribution network?*

# Problem framing

**Shifting the Dimensions of Risk**

The Foundation (Sangroula et al., 2026)

1. Physical
2. Environmental
3. Operational

**Our Approach**

- The Challenge: Missing or inaccessible structural physical pipeline data.
- The Solution: Pivot the problem structure to exploit the vast availability of Open Data in the remaining two dimensions.
- Focus Area: Building a flexible risk framework driven entirely by Environmental and Operational indicators.

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**Shifting the Dimensions of Risk**

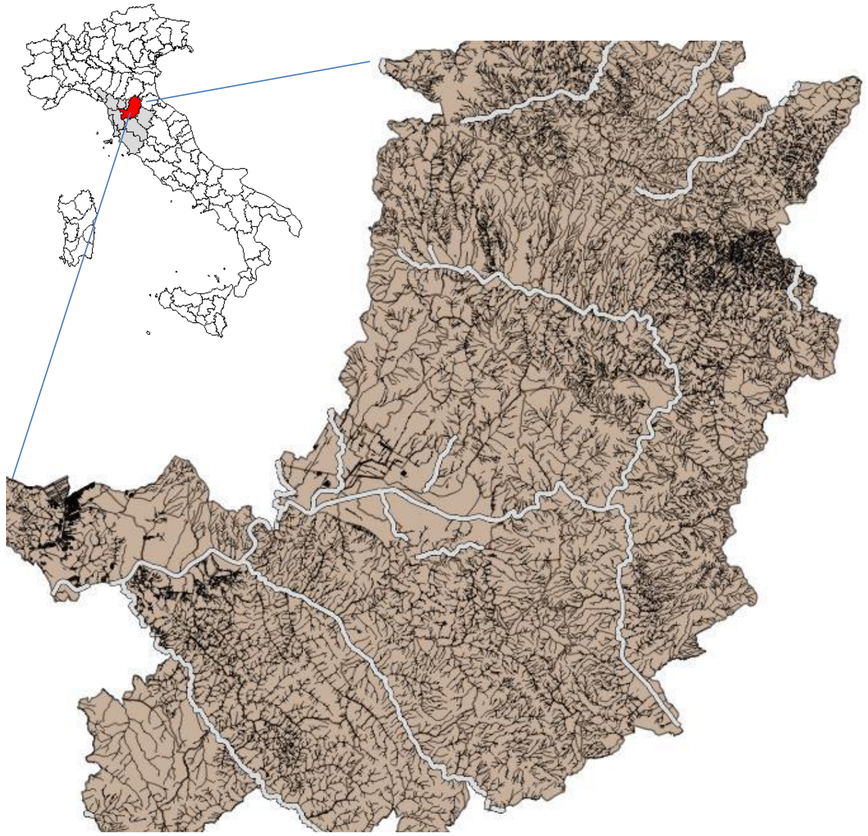
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Florence Province



**Case study & geospatial scope**

- Province of Florence (Firenze) initial phase.
- Initial scope manageable and controlled.
- Designed to scale the methodology across the Italian territory.
- Standardized Uber H3 grid
- Instead of mapping specific points we map zones
- If a pipeline is inside the zone, how hazardous would it be?

# Research question & objectives

**Main Research question**

- How can open data be leveraged to identify and analyze inefficiencies in water transportation systems through pipelines?

- How can multimodal data integration enhance the detection and analysis of inefficiencies in pipeline-based water transportation systems?

- How can existing patent solutions be utilized to address the inefficiencies in pipeline water transportation systems?



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**Secondary objectives**

- To identify and catalog diverse repositories of open data relevant to the inefficiencies of the water transportation system.

**Secondary objectives**

- To identify and catalog patent lists for their respective patent solutions relevant to the water transportation system.

**Secondary objectives**

- To design a data-driven framework that utilizes integrated insights to support the prioritization of infrastructure interventions and the optimization of resource allocation.





Methodology framing

*How can we pair causes with solutions?*

# Approach

## Vulnerability layer & Mitigation layer

**Vulnerability layer**

- Compiles all collected spatial layers into a single, unified map for every individual H3 hexagon cell.
- Output the hexagons which contain causal chains activate through environmental stressors;
- It breaks down the reasons why a specific hexagon is risky (e.g., High temperature in Inlet, cause, expansion).



Methodology framing

*How can we pair causes with solutions?*

# Approach

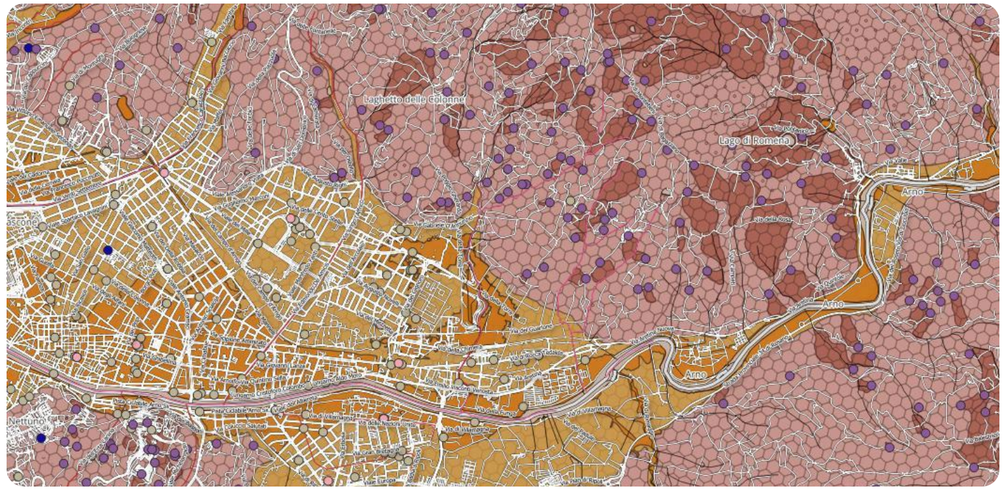
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**Vulnerability layer**

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- It breaks down the reasons why a specific hexagon is risky (e.g., High temperature, high rain, elevated seismic activity).

**Mitigation layer**

- The system ingests the critical values of the stressors generated by the vulnerability layer.
- Rather than giving generic warnings, it cross-references the hazard profile of the hexagon (e.g., high winds or steep slope).
- The system flags and shows specific **chains from patents** tailored display, why this would be a risk area, with cause and effects.



# Data sources

*Multimodal, reliable and open access*

### Environmental risk sites and Patent data

Programs like **SIR** from **Regione toscana**, **IdroGEO** from **Isprambiente**, **CFTI landslides** from the national institute of geophysics and vulcanology (**INGV**) and **Copernicus crops data.**

Patent data enhances technical knowledge about the physical factors that affect the pipeline systems at hand

### Preliminary exploration EUROSTAT & ISTAT

### Dati.gov.it and ARPA entities

### Open data regional sites




Main sources by content

- Official National Hazard Inventories for landslides, floods, seismicity and wildfires
- Decades of localized rain , temperature, humidity and wind gust history
- Communal income and population census
- Pedological soil databases, sentinel-1 soil displacement points
- Streets,  infrastructural assets,  waterways (OSM)
- Pendolarismo, daily commutters between communes
- Natural disasters interventions
- High-resolution DTMs

# Patents retrieval

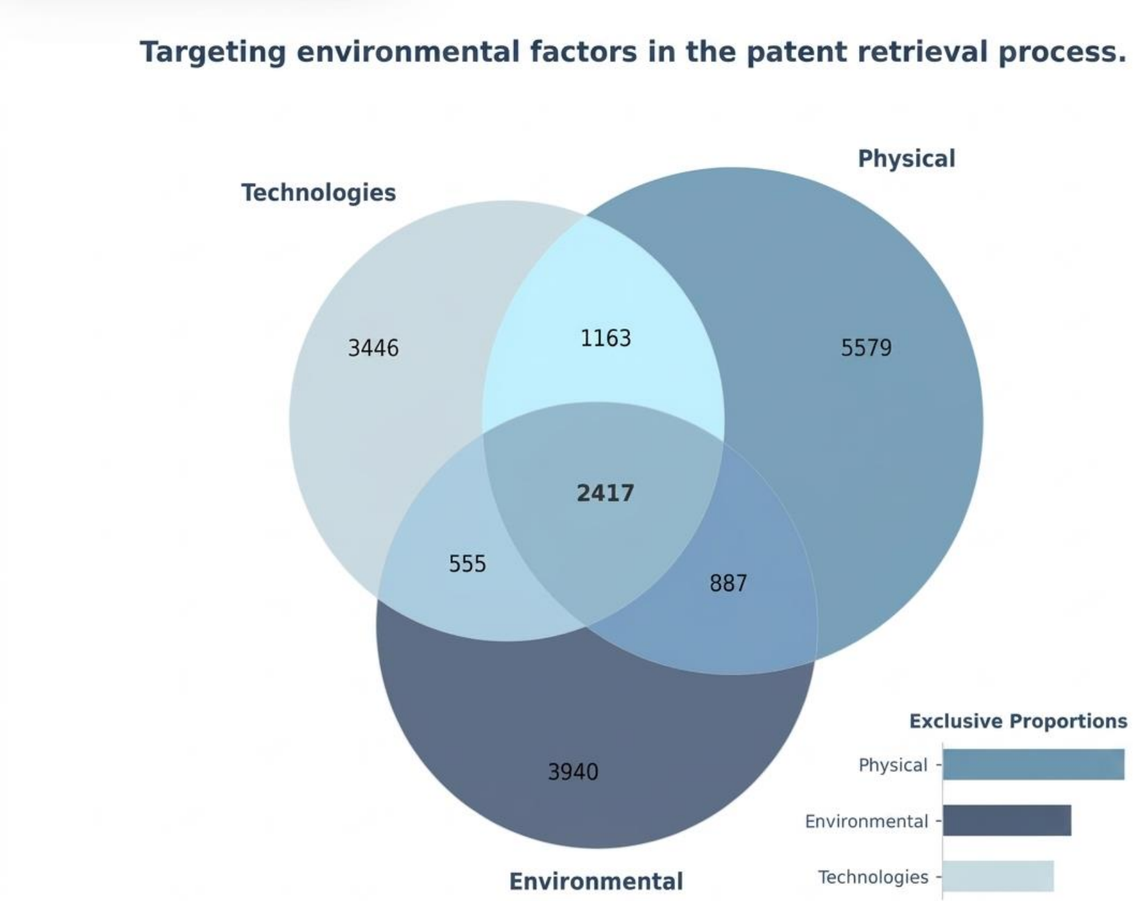
## Retrieval strategy

Extracted a comprehensive macro-pool of global patents targeted at:

- Water distribution networks
- Bulk water transportation
- Sewage and wastewater infrastructure

## The classification dimensions

- **Physical: Structural pipeline faults, joints, reinforcements, and physical assets.**
- **Environmental: Soil stabilization, erosion control, and external terrain-defense solutions.**
- **Technologies**: Digital systems, monitoring tools, smart valves, and software layers.



# Methodology

*Creating a framework for the causal chain solutions*

Data cube structure

Generate a schema that allows the system to assess the risk with the contextual information of the static properties mainly environmental conditions, plus the situational conditions that change in, weekly or monthly basis, fused with the patent data to form causal chains that can be related to environmental stressors.



**Causes**

- Technical components
- Physical nodes
- Pipelines
- Sensors
- Pumps

**Conditions**

- Environment attributes
- Zonal context
- Temperature
- Water levels
- Rain

**Effects**

- Failures
- Leakages
- Oxidation
- Blockage
- Contamination

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